

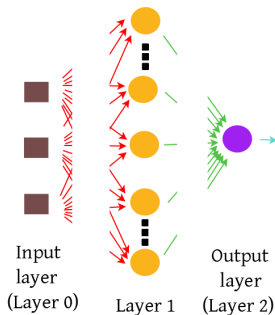
Deep Learning - Theory and Practice

IE 643
Lecture 10

September 1, 2025.

- 1 Popular examples of MLP
 - MLP with Single Hidden Layer
 - Encoder-Decoder Architecture

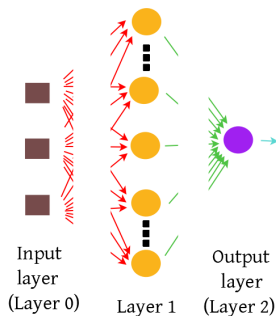
MLP with Single Hidden Layer: Approximation Properties



In the MLP with single hidden layer, consider:

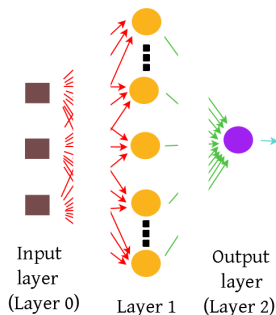
- d neurons in input layer accepting inputs from $[0, 1]^d$.
- N neurons in hidden layer.
- Single neuron in the output layer.

MLP with Single Hidden Layer: Approximation Properties



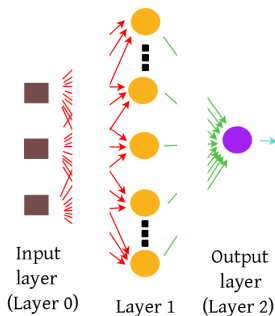
- All hidden layer neurons have sigmoidal activations.

MLP with Single Hidden Layer: Approximation Properties



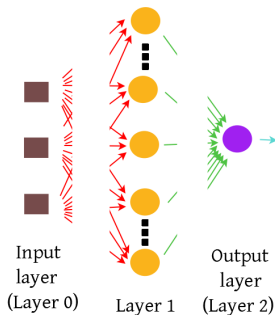
- All hidden layer neurons have sigmoidal activations.
 - ▶ **Recall:** A sigmoidal activation $\sigma : \mathbb{R} \rightarrow (0, 1)$ is a monotonically increasing continuous function with the property $\sigma(z) \rightarrow 0$ as $z \rightarrow -\infty$ and $\sigma(z) \rightarrow 1$ as $z \rightarrow +\infty$.

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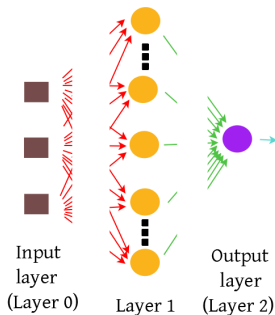
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- The output layer neuron has linear activation function.

MLP with Single Hidden Layer: Approximation Properties



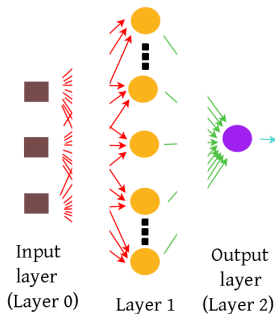
- All hidden layer neurons have sigmoidal activations.
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- The output layer neuron has linear activation function.
 - ▶ **Recall:** A linear activation function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is given by $\phi(z) = z$.

MLP with Single Hidden Layer: Approximation Properties



- Weights connecting the input layer neurons to the j -th neuron in the hidden layer are collected into the vector w_j and the associated bias be b_j .
- Weight connecting the j -th neuron in hidden layer to the output layer neuron is denoted by α_j .

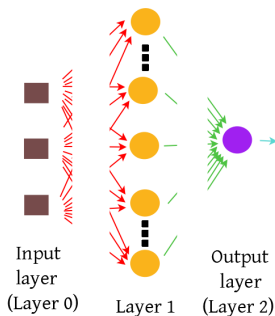
MLP with Single Hidden Layer: Approximation Properties



- Then for an input $x \in [0, 1]^d$ the prediction or last layer output can be represented as:

$$\hat{y} = G(x) = \sum_{j=1}^N \alpha_j \sigma(w_j^\top x + b_j).$$

MLP with Single Hidden Layer: Approximation Properties

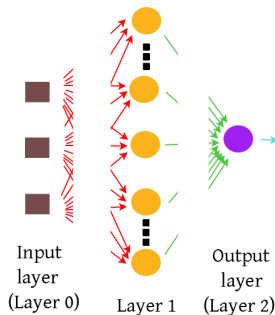


From a famous result of [Cybenko \(1989\)](#)[†] we have:

Let $\mathcal{C}([0, 1]^d)$ denote the set of continuous functions over $[0, 1]^d$ and let $\epsilon > 0$. Then for any $f \in \mathcal{C}([0, 1]^d)$, there is a sum of the form $G(x) = \sum_{j=1}^N \alpha_j \sigma(w_j^\top x + b_j)$ such that $|G(x) - f(x)| < \epsilon, \forall x \in [0, 1]^d$.

[†] G. Cybenko, Approximations by Superpositions of a Sigmoidal Function, Math. Control Signals Systems, 1989.

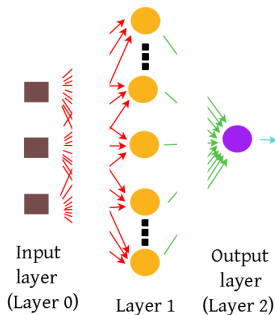
MLP with Single Hidden Layer: Approximation Properties



The result of [Cybenko \(1989\)](#) implies:

- Single hidden layer networks where hidden layer neurons have sigmoidal activation functions and an output neuron with linear activation can approximate any continuous function over $[0, 1]^d$ to any arbitrary precision $\epsilon > 0$.

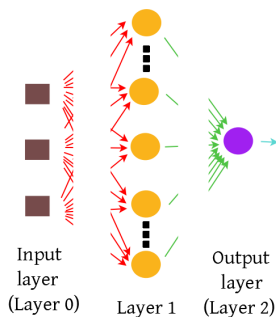
MLP with Single Hidden Layer: Approximation Properties



The result of [Cybenko \(1989\)](#) implies:

- Single hidden layer networks where hidden layer neurons has sigmoidal activation functions and an output neuron with linear activation can approximate any continuous function over $[0, 1]^d$ to any arbitrarily precision $\epsilon > 0$.
- This result is about the **approximation capability** of single hidden layer networks.

MLP with Single Hidden Layer: Approximation Properties



- **Caveat:** How many neurons N do we require in the hidden layer to achieve the approximation?